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ISE-503 Spring 2026: Health Analytics

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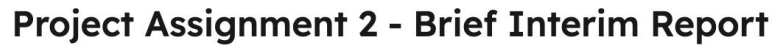
Title	Points
1 Health Problem Context	10.0 pts
1.1 Introduction / background P1 ✖	5.0 pts
1.2 Motivation P1 ✖	5.0 pts
2 Health Analytics Framework	20.0 pts
2.1 Objective and research question(s) P1 ✖	7.0 pts
2.2 Target Population P1 ✖	3.0 pts

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P1 x

P1 x

P1 x

P2 *

P2 x

P2 x

2.1 Statement of how 6.0 pts

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Project Assignment 2 - Brief Interim Report

Assign Questions and Pages

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3 Expected contribution of the approach	6.0 pts	
3.1 Statement of how modeling approach can contribute to addressing the health problem	6.0 pts	
4 Data Source and Description	10.0 pts	
4.1 Verifiable data source and reference	3.0 pts	
4.2 Dataset description	4.0 pts	
4.3 Suitability for answering the question	3.0 pts	
5 Methods and Analytic Pipeline	15.0 pts	
5.1 Overview of analytic approaches	10.0 pts	

ISE 503: Health Analytics

Project 2 - Brief Interim Report

Practicing High-Level Health and Patient Safety: You Receive High-Quality Data (Questions & Answers) - 10/10/2025

1

Q1.1 x

Q1.2 x

Q2.1 x

Q2.2 x

Q2.3 x

Decision-Making Problem and the Case

The primary use case in this problem is to generate a list of recommendations to help with the decision-making process. The goal is to identify the most important factors that influence the decision-making process. The goal is to identify the most important factors that influence the decision-making process. The goal is to identify the most important factors that influence the decision-making process.

2

Q2.4 x

Q2.5 x

Q2.6 x

Q3.1 x

Q4.1 x

Expected Contribution

The project will provide an independent, interdisciplinary perspective on the current state of the field. The project will provide an independent, interdisciplinary perspective on the current state of the field. The project will provide an independent, interdisciplinary perspective on the current state of the field. The project will provide an independent, interdisciplinary perspective on the current state of the field.

3

References

Chen, Y., He, Y., & He, Y. (2023). The impact of the COVID-19 pandemic on the health care system. *Journal of the American Medical Association*, 329(10), 900-901.

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Project Assignment 2 - Brief Interim Report

Assign Questions and Pages

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Select questions and pages to indicate where your responses are located. Use **esc** to deselect all items and hold **shift** to select multiple questions. Drag and drop pages to reorder them, or select a page and use the arrow keys on the keyboard.

5.1 Overview of analytic approaches	10.0 pts
5.2 Justification for choice of approaches including appropriateness for objective/data/audience	5.0 pts
6 Initial Results and Next Steps	15.0 pts
6.1 EDA	5.0 pts
6.2 Initial modeling	5.0 pts
6.3 Interpretation of findings	2.5 pts
6.4 Next steps	2.5 pts
7 Challenges	5.0 pts
7.1 Identification of challenges	2.5 pts

ISE-503 Health Analytics

Project 2 - Brief Interim Report

Practicing High-Level Problem Solving: You Receive High-Quality Data (Questions & Answers) - 10/16/2025

1. Introduction

2. Analytical Approach and Design

3. Initial Results and Next Steps

4. Challenges

Q1.1 x

Q1.2 x

Q2.1 x

Q2.2 x

Q2.3 x

Q2.4 x

Q2.5 x

Q2.6 x

Q3.1 x

Q4.1 x

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7 Challenges	5.0 pts
7.1 Identification of challenges	2.5 pts
7.2 Proposed solutions	2.5 pts
8 Appendix A	5.0 pts
8.1 Display items referred to in main text	2.5 pts
8.2 Display items presentation-ready format (not copy-pasted)	2.5 pts
9 Appendix B: Code	5.0 pts
10 Writing Quality and Formatting	9.0 pts
10.1 Clarity	3.0 pts
10.2 Organization	3.0 pts
10.3 Adherence to	3.0 pts

ISE 503: Health Analytics

Project 2 - Brief Interim Report

Presenting High-Risk Patients Before They Become High-Cost

David Chen & Sophia Lee | 10/10/2025

1. Introduction

Healthcare cost containment is one of the central challenges in U.S. health policy. Identifying high-risk patients early is critical for reducing costs and improving outcomes. This report presents a data-driven analysis of patient data from the ICD-9-CM database, focusing on patients with high-risk diagnoses and procedures. The goal is to identify high-risk patients early and intervene before costs escalate.

2. Data Analysis and Findings

The analysis of the ICD-9-CM database reveals several key findings. First, patients with high-risk diagnoses (e.g., heart failure, chronic obstructive pulmonary disease) are more likely to have high-cost procedures. Second, patients with high-risk procedures (e.g., heart surgery, organ transplant) are more likely to have high-cost diagnoses. Third, patients with high-risk diagnoses and procedures are more likely to have high-cost stays. These findings suggest that identifying high-risk patients early and intervening before costs escalate is a critical strategy for reducing healthcare costs.

3. Conclusion

This report provides a data-driven analysis of patient data from the ICD-9-CM database, focusing on patients with high-risk diagnoses and procedures. The findings suggest that identifying high-risk patients early and intervening before costs escalate is a critical strategy for reducing healthcare costs.

Decision-Making Problem and the Case

The primary use case in this problem is to identify high-risk patients early and intervene before costs escalate. This is a complex problem that requires a deep understanding of the data and the ability to make decisions based on the data. The case involves a patient with a high-risk diagnosis and a high-risk procedure, and the goal is to identify the patient early and intervene before costs escalate.

Primary Evaluation Focus

The primary evaluation focus is on the patient's diagnosis and procedure. The goal is to identify high-risk patients early and intervene before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

Problem-Solving Strategy

The problem-solving strategy involves identifying high-risk patients early and intervening before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

2. Analytical Approach and Data

The analytical approach involves identifying high-risk patients early and intervening before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

Feature Engineering

Feature engineering involves identifying high-risk patients early and intervening before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

Model Development

Model development involves identifying high-risk patients early and intervening before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

Model Evaluation

Model evaluation involves identifying high-risk patients early and intervening before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

Model Deployment

Model deployment involves identifying high-risk patients early and intervening before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

3. Results and Next Steps

Current Status

The current status of the project is that the data has been analyzed and the findings have been presented. The next steps are to identify high-risk patients early and intervene before costs escalate.

Next Steps

The next steps are to identify high-risk patients early and intervene before costs escalate. This requires a deep understanding of the data and the ability to make decisions based on the data.

4. Challenges

One challenge is the complexity of the data. The data is large and complex, and it is difficult to analyze. Another challenge is the need for a deep understanding of the data and the ability to make decisions based on the data.

Another challenge is the need for a deep understanding of the data and the ability to make decisions based on the data.

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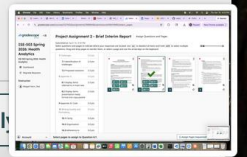
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7.1 Identification of challenges	2.5 pts
7.2 Proposed solutions	2.5 pts
8 Appendix A	5.0 pts
8.1 Display items referred to in main text	2.5 pts
8.2 Display items presentation-ready format (not copy-pasted)	2.5 pts
9 Appendix B: Code	5.0 pts
10 Writing Quality and Formatting	9.0 pts
10.1 Clarity	3.0 pts
10.2 Organization	3.0 pts
10.3 Adherence to formatting guidelines	3.0 pts

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